Assignment No. 2

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Q1. Given sentence: “The bank will not open until the river recedes.”

a) Identify and explain the ambiguity

Ans: The ambiguous word in the sentence is **“bank.”**

- **Bank (financial institution):** A financial place where people deposit, keep, or withdraw money.
- **Bank (river bank):** The sloping land directly beside a body of water or river.

Here, the correct meaning intended by context is **river bank**, because the sentence explicitly refers to the river receding.

Type: Lexical ambiguity / Word Sense Ambiguity.

b) How WordNet or BabelNet can help

Ans: WordNet lists different distinct meanings (known as synsets) of the word *bank* and provides semantic relationships (hypernyms, hyponyms) between words and their meanings.

BabelNet combines WordNet-like taxonomic information with multilingual encyclopedic knowledge from sources like Wikipedia.

The automated system can compare surrounding contextual words such as “river” and “recedes” with the glosses and semantic relations of the possible senses of *bank*. By calculating contextual overlap, it selects **bank = land beside a river**.

c) Why automatic systems may still fail

Ans: Automatic NLP systems can still fail due to the following factors:

- Context may be too short, sparse, or ambiguous to provide enough evidence.
- A word can have several fine-grained or closely related senses that are hard to distinguish.
- Dictionaries and knowledge bases may lack sufficient contextual or domain-specific coverage.
- Complex interpretation often requires real-world common-sense knowledge.
- Different specialized domains use identical words in highly specific, unconventional ways.

Q2. Naïve Bayes for Word Sense Disambiguation

a) How Naïve Bayes is applied to WSD

Ans: Naïve Bayes classifies a target ambiguous word into one of its possible pre-defined senses by evaluating surrounding context words. For example, for **"bank"**:

- **Sense 1 (S1):** Financial institution
- **Sense 2 (S2):** River side

The Bayes classifier calculates the posterior probability for each sense given the context features:

$$P(Sense \mid Context) \propto P(Sense) \times \prod P(Word_i \mid Sense)$$

The sense yielding the highest calculated probability is assigned. For instance, in *"The fisherman sat on the bank of the river,"* words like *fisherman*, *river*, and *sat* strongly maximize the probability for the river-bank sense.

Q2. Naïve Bayes for Word Sense Disambiguation

a) How Naïve Bayes is applied to WSD

Naïve Bayes classifies a target word into one of its possible senses using the surrounding context.

For example, for "bank":

Sense 1 → financial institution

Sense 2 → river side

The classifier calculates:

The sense having the highest probability is selected.

Example:

"The fisherman sat on the bank of the river."

Words like fisherman, river, sat strongly support the river-bank sense.

b) Limitations compared to Lesk's approach

Feature / Dimension	Naïve Bayes Approach	Lesk's Approach
Training Data	Requires supervised sense-annotated training data.	Mainly knowledge/dictionary based (unsupervised).
Learning Method	Learns probabilistic weights from training examples.	Uses dictionary definitions and gloss overlaps.
Core Mechanism	Assumes context features are conditionally independent.	Compares semantic overlap between glosses and context.
Performance Basis	Performance depends heavily on quality of training data.	Can function effectively with limited training data.
Unseen Contexts	May perform poorly on out-of-domain/unseen context.	Leverages dictionary definitions dynamically.
Domain Adaptability	Requires domain-specific re-training.	Uses general dictionary overlap definitions

Q3. Given sentence: “John met Alex after he finished the project.”

a) Identify the anaphora and explain the difficulty

Ans: The anaphoric expression is the pronoun **“he.”**

The candidate antecedent entities are:

- **John** (Male proper noun)
- **Alex** (Male proper noun)

Difficulty: The sentence structure does not explicitly declare which entity “he” refers to. Both candidates share matching gender and number properties, creating reference ambiguity that syntactic cues alone cannot conclusively resolve.

b) How would Hobbs' algorithm resolve “he”?

Ans: Hobbs' algorithm operates strictly on hierarchical syntactic parse trees using standard breadth-first search procedures:

1. Parse the input sentence into a structural syntactic parse tree.
2. Locate the target pronoun node (*“he”*).
3. Ascend the parse tree to the first NP or S node dominating the pronoun.
4. Traverse left of the path searching for candidate Noun Phrases (NPs).
5. Filter candidates matching agreement constraints (gender: Male, number: Singular).
6. Select the first matching candidate encountered during search order.

For this sentence, because both *John* and *Alex* fulfill standard agreement constraints, standard parse tree traversal order generally favors the matrix subject (*John*), though syntactic rules alone remain ambiguous without semantic verification.

c) Real-world scenario where syntax alone fails

Ans: “The trophy did not fit in the suitcase because it was too large.”

Syntactically, “it” could refer to either trophy or suitcase.

But common sense tells us that the trophy is more likely to be too large to fit inside the suitcase.

Therefore, syntactic constraints alone are not always sufficient; real-world knowledge and semantics are also needed.